

Predictive Maintenance & Condition Monitoring

E-Bike Manufacturing — Enhanced MES Reference

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This dataset carries a live condition-monitoring layer: per-machine sensors, a high-rate vibration analysis stream, and a predictive-maintenance model that raises failure predictions with confidence, remaining-useful-life (RUL) and priority. This guide explains the signals and the alert-triage workflow.

1. Sensor Network

Instrumentation is concentrated on the three frame-welding machines (Fra-10/11/12), which are the plant's lowest-OEE, highest-vibration assets.

Sensor Type	Unit	Count	Sampling Rate	Placement
Temperature	°C	3	1 Hz	Motor
Vibration	mm/s	9	1000 Hz	Spindle, Table, Tool_Head

2. Vibration Analysis Stream

The VibrationAnalysis table holds **6,483 snapshots** combining vibration, thermal and process signals with a computed health score and anomaly flag.

- **Vibration:** RMS_Velocity (mm/s), Peak_Acceleration, Frequency_Dominant (Hz).
- **Thermal / load:** Temperature, Load_Current.
- **Process:** Spindle_Speed, Feed_Rate.
- **Derived:** Health_Score (0–1) and Anomaly_Flag (1 = flagged).

Current signature: mean RMS velocity 5.13 mm/s, peak acceleration up to 221.3, mean temperature 57.5°C, mean health score 0.49, with **6,465 of 6,483 snapshots flagged as anomalous**.

A mean health score around 0.5 with a very high anomaly rate indicates these welding spindles are running in a degraded band — consistent with their position at the bottom of the OEE table. They are the priority candidates for the actions in Section 4.

Machine	Mean Health	Anomalies	Snapshots
Machine Fra-10	0.49	2,155	2,161
Machine Fra-11	0.49	2,155	2,161
Machine Fra-12	0.49	2,155	2,161

3. Operating Limits & Alarm Thresholds

Each condition-monitoring snapshot is evaluated against fixed alarm thresholds. A reading that reaches the Warning band is flagged (Anomaly_Flag = 1); a reading that reaches the Critical band requires the machine to be stopped and a maintenance ticket raised. These documented limits are the reference against which the recorded VibrationAnalysis operating values are compared.

Signal	Normal	Warning	Critical / stop
Motor temperature (°C)	< 70	70 – 85	≥ 85
RMS vibration velocity (mm/s)	< 7.1	7.1 – 18.0	> 18.0
Peak acceleration (g)	< 4.0	4.0 – 8.0	> 8.0

Signal	Normal	Warning	Critical / stop
Health score (0–1)	≥ 0.70	0.40 – 0.69	< 0.40

- **Motor over-temperature** — the 85°C Critical limit protects the drive motor; sustained operation above it accelerates bearing and winding degradation and must trigger a stop.
- **Vibration bands** follow ISO 10816-style severity zones for medium machines.
- **Anomaly Flag** is set whenever any signal reaches its Warning or Critical band; Health_Score is the combined 0–1 index used for trending.

Limit checks the retrieval system should support

- Compare a machine's recorded VibrationAnalysis readings (temperature, RMS velocity) against the documented limits above to decide whether it operated within tolerance during a given production run.
- Cross-reference a Critical breach with the predictive-maintenance queue and the relevant SOP to choose the corrective action.

4. Predictive-Maintenance Alerts

The model writes predictions to PredictiveMaintenance with a failure mode, predicted failure date, confidence, RUL, recommended action, priority and status.

Failure Mode	Alerts	Avg Confidence	Avg RUL (h)
belt wear	16	0.75	344
lubrication failure	14	0.80	650
tool wear	12	0.79	533
spindle imbalance	12	0.76	738
bearing failure	12	0.80	697
motor fault	11	0.74	734
vibration excessive	10	0.75	583
overheating	6	0.73	669

Alert inventory by priority and status

Priority	active	false_positive	resolved	Total
Critical	1	2	—	3
High	7	4	5	16
Medium	9	23	10	42
Low	17	7	8	32

Note the large share of 'false_positive' alerts — the triage workflow below exists specifically to separate these from genuine degradation before spending a maintenance window.

5. Alert Triage Workflow

- Rank** the active queue by Priority (critical → high → medium → low) then by shortest RUL.
- Corroborate** — cross-check the alert against VibrationAnalysis health/anomaly trend and recent OEE availability for the same machine.
- Classify** — if signals do not support the prediction, mark Status = false_positive and feed back to the model. Otherwise keep it active.
- Schedule** — for confirmed alerts, apply the recommended action within the RUL horizon, ideally in the next planned window (Downtimes 'Scheduled Maintenance').
- Resolve** — after the work, set Status = resolved and update the machine's Last/Next maintenance dates.

Escalation

- Critical + active → same-shift engineering review
- High + active with RUL < 168 h (one week) → schedule within the week
- Repeated false positives on one mode/machine → retune the model threshold